CUSTOMER CHURN ANALYSIS

End-to-End Data Analytics Project Report

SQL • Python • SQLite • Pandas • NumPy • Matplotlib • Seaborn

Portfolio-ready report based on the supplied project database and Jupyter Notebook.

1. Executive Summary

This project analyzes customer churn for a subscription-based OTT business using a relational SQLite database and Python. The workflow integrates customer, subscription and support information to identify churn patterns, quantify revenue exposure, segment customers using the existing churn score, and translate findings into retention actions.

The supplied dataset contains 21 customers. Overall churn is 28.6% and retention is 71.4%. Monthly-contract churn is 55.6% versus 8.3% for annual contracts.

2. Business Problem & Objectives

The analysis focuses on who is churning, why churn is happening, when churn is concentrated, and which customers should be prioritized.

• Measure churn and retention.

• Compare churn across plan and contract types.

• Identify geographic and monthly churn concentrations.

• Quantify revenue exposed to churn.

• Segment customers using the existing churn_score.

• Examine support escalation as a churn-related signal.

• Translate findings into practical retention actions.

3. Technology Stack & Workflow

• SQLite and SQL for relational extraction.

• Python with pandas and NumPy for transformation and analysis.

• Matplotlib and Seaborn for visualization.

• Group-by analysis, pivot tables, KPI calculations and correlation analysis.

• Jupyter Notebook for reproducible analysis.

4. Data Architecture

Table	Purpose	Key fields
db_customer	Customer attributes	customerid, country, state, gender, dob
db_subscription	Subscription & commercial data	plan_type, contract_type, monthly_charges, cltv, churn_score, cancellation_date
db_support	Support interactions	complaint_date, escalations, csat_score

5. Data Preparation & Feature Engineering

• Renamed and removed unused fields.

• Converted date fields and standardized categories.

• Filled missing country values using state-to-country mapping.

• Created churn_flag from cancellation_date.

• Calculated complaint_count and tenure_days.

• Merged all three tables on customerid.

• Created Low / Medium / High risk bands from churn_score.

6. KPI Dashboard

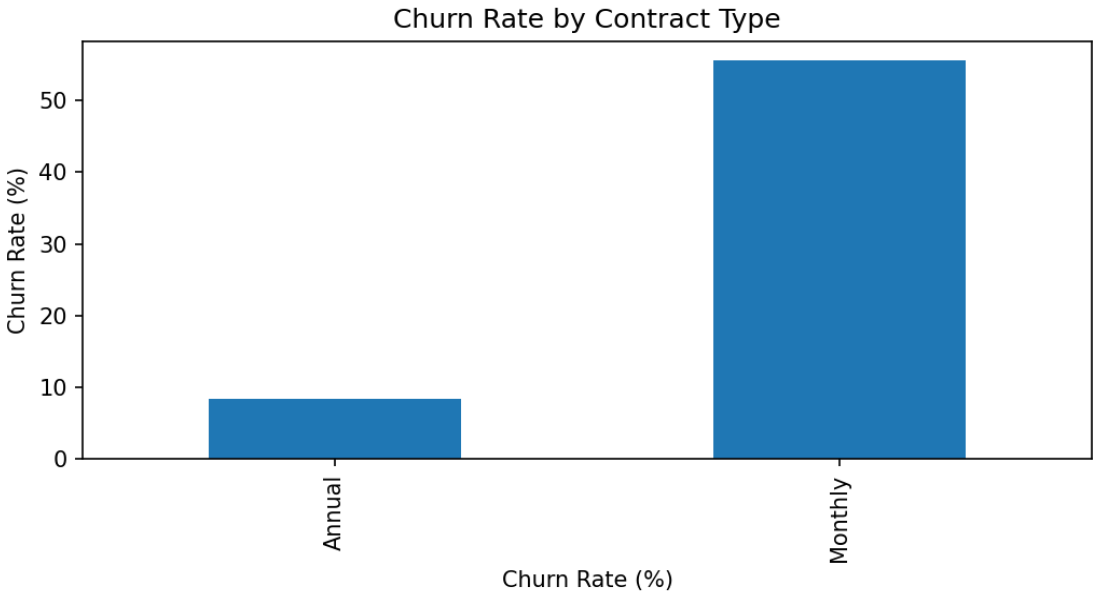
KPI	Result
Customers	21
Churn rate	28.6%
Retention rate	71.4%
ARPU	■18.85
Average tenure	1,560 days
Total monthly charges	■395.79K
Revenue at risk	■73.94K
Revenue exposed	18.7%
Escalation rate	19.0%
Escalation–churn correlation	0.47

Monetary scale follows the supplied project dataset/reporting logic.

7. Churn Analysis

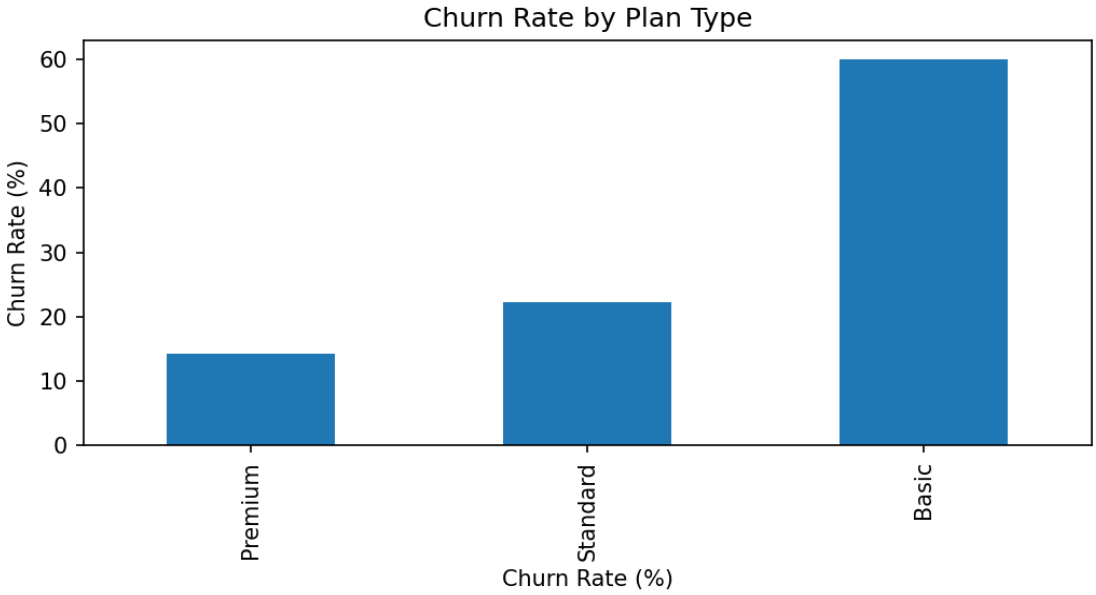
7.1 Contract Type

Monthly-contract churn is 55.6% versus 8.3% for annual contracts.



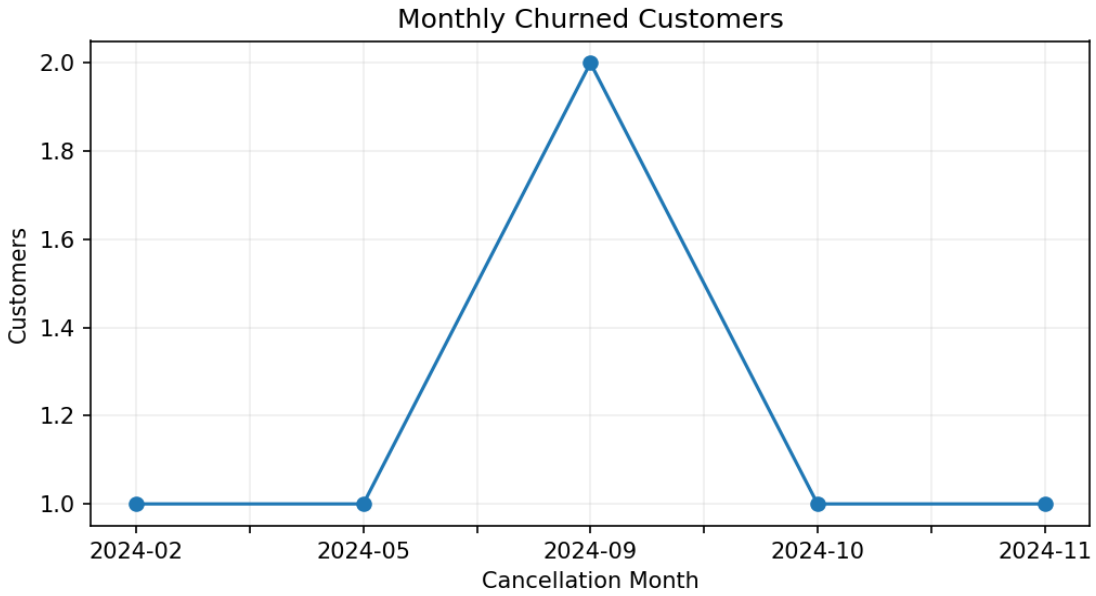
7.2 Plan Type

Basic has the highest observed churn at 60.0%, followed by Standard and Premium.



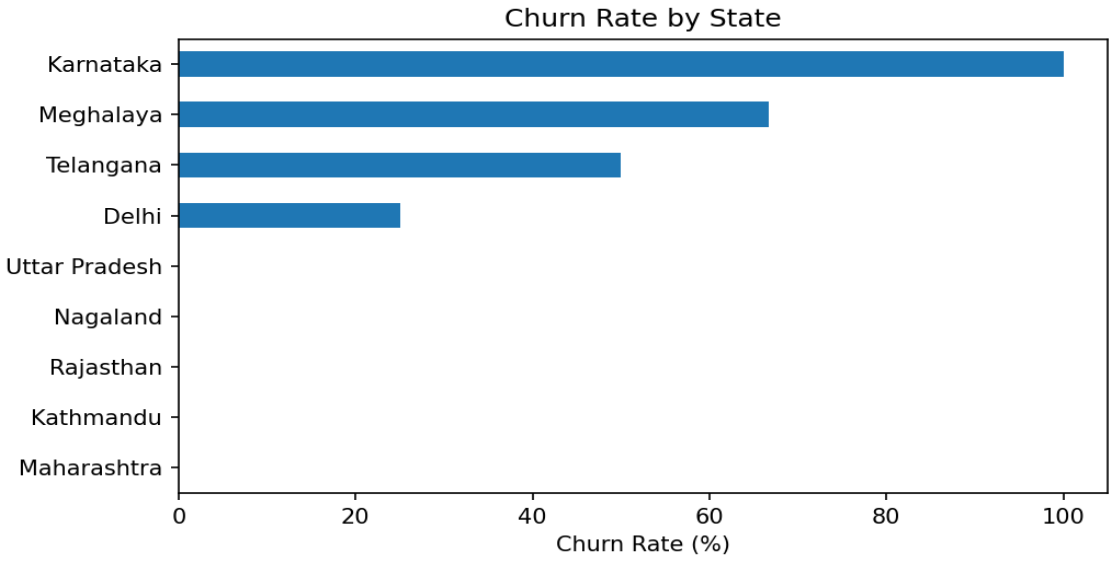
7.3 Monthly Churn

September 2024 has the highest monthly churn concentration, with two cancellations.



7.4 Geography

Karnataka has 100% churn in this sample (2 of 2 customers). Treat this as an investigation signal.



8. Cancellation Drivers

Cancellation reason	Customers
Switched to competitor	2
Too expensive	1
Not enough content	1
Poor streaming quality	1
Forgot to cancel trial	1

Observed reasons include competitor switching, price sensitivity, insufficient content, poor streaming quality and trial cancellation.

9. Risk Segmentation & Revenue Exposure

Customers with churn_score >= 70 are High Risk. Six customers fall into this segment and all six are churned in the supplied data. They represent ■73.94K in monthly charges and ■2,047 in CLTV.

Risk	Customers	Churned	Monthly	CLTV
Low	13	0	■286.87	■14,057
Medium	2	0	■34.98	■1,190
High	6	6	■73.94	■2,047

10. Support Intelligence

Escalation–churn correlation is approximately 0.47. This indicates association in the small sample, not causation. A larger longitudinal dataset is required for causal or predictive conclusions.

11. Business Recommendations

• Prioritize monthly-to-annual migration experiments.

• Investigate Basic-plan pricing, content and service quality.

• Investigate the Karnataka outlier through complaints, pricing and technical issues.

• Prioritize high-risk customers using churn score, CLTV, monthly charges and support history.

• Create targeted retention playbooks for cancellation reasons.

• Use support escalation as an early-warning signal and validate it with more data.

12. Limitations

• Only 21 customers are present.

• Analysis is descriptive/diagnostic and does not establish causation.

• churn_score is an existing dataset field; the notebook does not train a model to generate it.

• Support records are consolidated to the latest record after complaint counts are calculated.

• Geographic findings require validation on a larger dataset.

13. Presentation Storyline

• 1. Title & business problem

• 2. Dataset and data architecture

• 3. Data cleaning and feature engineering

• 4. KPI dashboard

• 5. Contract and plan churn

• 6. Geography and monthly trends

• 7. Risk segmentation and revenue at risk

• 8. Cancellation reasons and support

• 9. Recommendations

• 10. Limitations and conclusion

Suggested headline: “Monthly contracts are the highest-risk retention segment, while high-risk customers represent a concentrated and actionable revenue exposure.”

14. Conclusion

The project demonstrates an end-to-end analytics workflow from SQL extraction through Python transformation, KPI analysis, visualization, risk segmentation and business recommendations. The strongest observed signal is the large churn gap between monthly and annual contracts, supported by additional signals from plan type, geography, cancellation reasons and customer risk scores.